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CSD-380

### Assignment 1.3

DevOps is a software development philosophy that emphasizes collaboration between development and operations teams to deliver software more reliably and efficiently. While the term "DevOps" was not coined until 2009, its roots can be traced to several earlier movements that sought to improve software development processes. Three of the most influential predecessors of DevOps are the Lean Movement, the Agile Manifesto and the Continuous Delivery Movement. These approaches introduced principles that focused on reducing waste, improving collaboration and accelerating the delivery of value to customers. Understanding these historical influences provides valuable insight into how DevOps evolved into the widely adopted methodology used by organizations today.

#### **The Lean Movement**

The Lean Movement originated in the manufacturing industry, most notably through the Toyota Production System developed in Japan after World War II. Lean focuses on maximizing customer value while minimizing waste. Rather than concentrating solely on productivity, Lean seeks to improve the flow of work throughout an entire system.

Several Lean principles later became foundational concepts within DevOps. One such principle is the elimination of waste. Eliminating waste includes addressing unnecessary waiting, excessive handoffs and redundant work. Another important principle is continuous improvement. Rather than treating process improvement as a one-time project, Lean views continuous improvement as an ongoing responsibility shared by everyone within the organization.

The software industry began adopting Lean principles in the early 2000s through the Lean Software Development movement. Software teams recognized parallels in manufacturing systems and found that the development process contained the same types of waste and inefficiencies. Manual processes (and lack of automated ones), excessive documentation, lengthy approval processes and communication barriers often slowed the delivery of software. Lean encouraged organizations to streamline these processes and focus on delivering value to customers more efficiently. These concepts later became central goals of DevOps.

### **The Agile Manifesto**

Another major influence on DevOps was the Agile software development movement. In 2001, a group of seventeen software developers created what became known as the Agile Manifesto. The manifesto outlined four core values and twelve principles designed to improve software development practices.

The Agile Manifesto emphasized flexibility, adaptability and collaboration while it discouraged a rigid development process overly focused on documentation and contract negotiation. These principles represented a departure from traditional software development approaches such as the Waterfall model, which often relied on lengthy planning phases and infrequent releases.

Although Agile significantly improved development practices, many organizations discovered that development teams could create software faster than operations teams could deploy it. This mismatch created a bottleneck between development and operations departments. DevOps emerged in part as a solution to extend Agile principles beyond software development and into deployment, addressing infrastructure management and operational support.

## **The Continuous Delivery Movement**

The Continuous Delivery Movement provided another important foundation for DevOps. Continuous Delivery focuses on ensuring that software can be released to production quickly and safely. The movement gained significant attention through the work of Jez Humble and David Farley, particularly with the publication of their book *Continuous Delivery* in 2010.

Traditional software deployment processes were often highly manual and error-prone. Releases were infrequent and carried significant risk because organizations accumulated many changes before deploying them. Continuous Delivery sought to address these challenges by introducing automation throughout the software delivery pipeline. An important concept of continuous delivery is that code is in a deployable state at any time.

Key practices of Continuous Delivery include automated testing, automated builds, continuous integration, deployment pipelines, infrastructure as code and frequent releases of small changes. These practices reduce the likelihood of deployment failures and provide rapid feedback to development teams.

Continuous Delivery aligns closely with DevOps goals because it emphasizes automation and collaboration while prioritizing the rapid delivery of value. Many modern DevOps practices, including continuous integration and continuous deployment, evolved directly from the Continuous Delivery Movement.

## **Conclusion**

The development of DevOps was influenced by several important movements that transformed the software industry. The Lean Movement introduced principles focused on eliminating waste and continuously improving processes. The Agile Manifesto emphasized collaboration and adaptability during software development. The Continuous Delivery Movement expanded these ideas by promoting automation and reliable software releases. DevOps was built on a foundation of concepts from these movements. Today, DevOps is a foundational

approach in modern software development. Organizations and businesses employ DevOps principles to deliver quality software more efficiently while prioritizing communication and collaboration among teams and customers.

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